

Question ID: 79cf8505

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

A line segment that has a length of **115 centimeters (cm)** is divided into three parts. One part is **47 cm** long. The other two parts have lengths that are equal to each other. What is the length, in **cm**, of one of the other two parts of equal length?

Correct Answer: 34

Rationale

The correct answer is **34**. It's given that a line segment has a length of **115 cm** and is divided into three parts, where one part is **47 cm** long and the other two parts have lengths that are equal. If **x** represents the length, in cm, of each of the two parts of equal length, then the equation **$47 + x + x = 115$** , or **$47 + 2x = 115$** , represents this situation. Subtracting **47** from each side of this equation yields **$2x = 68$** . Dividing each side of this equation by **2** yields **$x = 34$** . Therefore, the length, in cm, of one of the two parts of equal length is **34**.